|INCH-POUND|

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SUPERSEDING
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MILITARY SPECIFICATION

# GRINDING MACHINES, CYLINDRICAL, EXTERNAL, CENTER TYPE UNIVERSAL, TRAVELING TABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This Specification covers electric motor driven, floor mounted, horizontal spindle, internal and external, cylindrical, universal grinding machines with equipment and accessories specified in section 3. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establish the Government's minimum requirements for each procurement.

1.2 Classification. The grinding machine shall be of the following sizes. The sizes to be furnished shall be as specified (see 3.5 and 6.2).

| Size | Rated swing diameter (inches) x distance between centers (inches) |
|---|---|
| 1020 | 10 x 20 |
| 1024 | 10 X 24 |
| 1036 | 10 X 36 |
| 1236 | 12 X 36 |
| 1248 | 12 X 48 |
| 1272 | 12 X 72 |

| Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. |
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| Size | Rated swing diameter (inches) x distance between centers (inches) |
|---|---|
| 1424 | 14 X 24 |
| 1436 | 14 X 36 |
| 1440 | 14 x 40 |
| 1448 | 14 X 48 |
| 1472 | 14 X 72 |
| 1496 | 14 X 96 |
| 1824 | 18 X 24 |
| 1836 | 18 X 36 |
| 1848 | 18 X 48 |
| 1872 | 18 X 72 |
| 1896 | 18 X 96 |
| 2424 | 24 X 24 |
| 2436 | 24 X 36 |
| 2448 | 24 X 48 |
| 2472 | 24 X 72 |
| 2496 | 24 X 96 |
| 3024 | 30 X 24 |
| 3036 | 30 X 36 |
| 3048 | 30 X 48 |
| 3072 | 30 X 72 |
| 3096 | 30 X 96 |

## 2. APPLICABLE DOCUMENTS.

### 2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.



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MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. Tolerances and Measuring methods for Unassembled Spur and Helical Gears. (Including metric equivalents) (DOD adopted)



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AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. (DOD adopted)

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues. (DOD adopted)

ANSI B5.9 - Spindle noses for Toolroom Lathes, Engine Lathes, Turret lathes, and Automatic Lathes. (DOD adopted)

ANSI B5.10 - Machine Tapers. (DOD adopted)

ANSI B7.1 - Safety requirements for the Use, Care, and Protection of Abrasive Wheels. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and for Shapes, Sizes, and Identification of Mounted Wheels. (DOD adopted)

ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN : Sales Department, 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA ICS 3 - Industrial Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)



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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection association, Batterymarch Park, Quincy, MA 02269-9101.)

NATIONAL FLUID POWER ASSOCIATION (NFPA)

NFPA/JIC T2.24.1- Hydraulic Fluid Power - Systems Standard for Stationary Industrial Machinery.

(Application for copies should be addressed to the National Fluid Power Association, 3333 N. Mayfair Road, Milwaukee, WI 53222)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3* REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete grinding machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The grinding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.0001 inch for the inch system, not larger than 0.003 mm for the metric system, or not larger than 0.0002 inch and 0.005 mm for the dual system of measurements. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a non-glare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero



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adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. Unless otherwise specified (see 6.2), a digital readout system shall be installed on the x and z axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation-of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.00005 inch. The system accuracy shall be 0.0001 inch. The repeatability shall be 0.00005 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input source fluctuation of ± 10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overdrive. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable from a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed to be readily accessible and easily read from-the operator’s normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of the installed cables shall prevent any interference with normal machine operation.



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3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured products are allowed under this specification.

3.2.3 Energy efficiency. The grinding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, electical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health retirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. All guards shall allow access to the guarded part(s).

3.2.6 Environmental protection. The machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. During the operation of the machine at its final location, it shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract regulating the emission of materials hazardous to the environment. Specific environmental protection requirements exceeding the established EPA and OSHA regulations as required herein shall be as specified by the procuring activity (see 6.2).

3.2.7 Lubrication. An automatic lubrication system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.7.1 Lubricants. Any greases, oils, or fluids used to initially charge the machine shall be free of contaminants known to be hazardous to human health.



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3.2.8 Interchangeability. All parts of the end item shall be manufactured to definite standards, dimensions, and tolerances that will provide the interchangeability of the respective replacement parts between end items of the same model without modification of the part or machine. All replacement parts shall be domestically available.

3.3 Construction. The grinding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. Any excess material used in such processes shall be thoroughly removed from the part(s) upon completion of the operation.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88 and AMA 2001-688 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. Drive train gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.



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3.3.8 Electromagmetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The grinding machine shall consist of not less than a bed and base, workhead, footstock, traversing table, swivel table, grinding wheelhead assembly, driving mechanisms and controls.

3.4.1 Bed and base. The bed and base shall be of a material of sufficient strength and rigidity for supporting and maintaining mutual component alignment as necessary for meeting performance, accuracy and alignment requirements herein. The bed shall have guideways for the support and precision movement of the traversing table along the longitudinal (Z) axis and the grinding wheel base assembly along the cross (X) axis. Worktable guideways shall be of such length that the center of gravity of the headstock and footstock is over the bed at all times during normal grinding operations. Leveling screws and floor mounts shall be provided around the periphery of the base. The hydraulic and coolant reservoirs may be housed in the base or provided as separate external units. When applicable, an access door(s) or removable door(s) shall be provided for servicing components housed within the base.

3.4.2 Workhead. The workhead shall house the spindle and its components and shall be longitudinally adjustable along the swivel table. Means for positive clamping of the workhead to the swivel table shall be provided. Unless otherwise specified (see 6.2), the workhead shall be arranged for both live and dead center grinding with provisions for rapid changeover to either operation. Necessary equipment (drive plates, pins, centers, etc.) for both live and dead center grinding shall be provided. Spindle noses and tapers shall conform to ANSI B5.9 and ANSI B5.10. The spindle shall be mounted in precision antifriction bearings with continuous lubrication or permanently lubricated and sealed for life. The workhead shall swivel on its base not less than 90 toward the wheelhead to facilitate angle and face grinding. The workhead shall have means to indicate swivel position in degrees and provisions for locking the workhead in the desired position within the swivel range.

3.4.3 Footstock. The footstock shall be longitudinally adjustable along the swivel table. Means for positive clamping to the swivel table after adjustment shall be provided. A handwheel, lever or other means shall be provided for adjusting spindle movement. The spring tension of the footstock center shall be adjustable to ensure proper pressure for the weight and size of the workpiece. Unless otherwise (see 6.2), the internal taper of the footstock center shall be the manufacturer’s standard. All moving parts shall be shielded against grit and coolant.

3.4.4 Traversing table. The traversing table shall support a swivel table and its components and shall traverse longitudinally (Z axis) on the guideways of the bed. Unless otherwise specified (see 6.2), table traversing shall be by power with means for manual control. The length of the traversing table travel shall be fully adjustable. The length of table stroke, with automatic



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reversal, shall be infinitely variable from a minimum of not more than 1/4 inch to the maximum stroke of the machine. Means shall be provided to prevent over-travel of the table in the event of a malfunction in the automatic table reversing mechanism. Manual traversing of the table shall be by handwheel, lever, or similar device. The table shall have an adjustable dwell control (O to not less than 2-1/2 seconds) for interrupting table movement at each table reversal. The table reversing mechanism, at all constant traverse rates, shall repeat within an accuracy of 0.004 inch at each table reversal.

3.4.5 Swivel table. The swivel table shall be mounted on the traversing table and shall be capable of swiveling in the horizontal plane for grinding included angles and tapers. Swivel adjustment shall be sufficient to grind a 13 degree included angle taper. A scale , indicating taper, graduated in both inches per foot and degrees of angle shall be provided. Means shall be provided for locking the swivel table in the desired position. When T-slots are provided for longitudinal adjustment of the headstock and footstock, they shall conform to ANSI B5.1.

3.4.5.1 Swivel table alignment device. Unless otherwise specified (see 6.2) the machine shall have a swivel alignment device which shall be operated by either mechanical or electronic means. The device shall allow rapid adjustment of the swivel table, in relation to the sliding table, to obtain and hold accurate parallelism of the work or for accurate angular setting for grinding taper.

3.4.6 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a wheelhead housing, spindle, and spindle drive mechanism. The assembly shall be mounted on guideways provided on the bed and shall traverse cross axially (X axis) to the table. The crossfeed range shall permit grinding from the stated maximum diameter to 0.000 diameter. A graduated scale shall be provided indicating swivel position in increments of not less than one degree. Means for locking the wheelhead in place after setting shall be supplied. The spindle shall be dynamically balanced and shall be supported by precision antifriction bearings designed to minimize axial and radial play. The spindle bearings shall be continuously lubricated or permanently lubricated and sealed for life.

3.4.6.1 Automatic feed. The machine shall have provisions for automatic infeed of the abrasive wheel at each table reversal, for traverse grinding operations. The arrangement shall include means for controlling infeed increments at desired reversals and for limiting the total infeed for the traverse grinding cycle. the wheelfeed mechanisms shall be actuated upon each reversal of the table’s longitudinal movement.

3.4.6.2 Manual infeed. A safety interlock shall prevent simultaneous operation of the automatic and the manual wheelhead infeed. Manual infeed shall be accomplished utilizing a handwheel or other device. The hand feed shall have both a coarse and fine feed with a scale or indicator graduated in accordance with 3.2.1.1.



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3.4.6.3 Power rapid positioning. When specified (see 6.2), the grinding wheelhead assembly shall have power rapid positioning. The design of the system shall not permit simultaneous engagement of manual feed and power positioning.

3.4.6.4 Internal grinding wheel unit. The grinding machine shall be equipped with an internal grinding unit consisting of a spindle, quills, and drive motor. The spindle shall be supported in precision antifriction bearings, permanently lubricated and sealed for life. The spindle shall be designed to accept various size quills to facilitate grinding different depths and to allow for the use of a variety of grinding wheel sizes.

3.4.7 Automatic infeed cycle. When specified (see 6.2), the machine shall have an automatic infeed cycle for plunge grinding. After set-up for the automatic infeed cycle the operator shall only be required to. load the workpiece, start the cycle and unload the workpiece. The automatic infeed cycle shall include, but not be limited to, the following functions:

a. Wheel advances at fast travel, headstock rotation starts, coolant flow starts.

b. Grinding feedrate automatically engages as wheel nears or contacts work (rate shall be adjustable).

c. Fine feed automatically engages as work nears finished size (rate shall be adjustable).

d. Spark-out. After the positive stop for the cross feed is reached, spark-out timing mechanism is actuated.

e. Wheelslide retracts rapidly, feed mechanism is reset to its original position, headstock rotation stops, coolant flow stops.

3.4.8 Drive mechanisms. The drive mechanisms for both the workhead and the wheelhead shall be by electric motor with belt and pulley arrangement. The mechanisms shall have means for adjusting belt tension unless belt tension is maintained automatically.

3.4.9 Wheel truing device. The machine shall be equipped with a wheel truing device that is arranged to dress the entire periphery of the grinding wheel. The device shall be complete with a commercial type diamond of not less than one (1) carat.

3.4.10 Grinding wheels. Unless otherwise specified (see 6.2), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI 07.1. All grinding wheels shall be properly guarded.

3.4.11 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a



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strainer which is easily removable for cleaning. An optional filtration system required in lieu of the manufacturer's standard system shall be as specified and fully described (see 6.2). Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.12 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of NFPA/JIC T2.24.1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level and provisions for draining.

3.4.13 Electrical system. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed for operation on dual 230/460-volt, 3-phase, 60-hertz (Hz) input power. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired for operation on the voltage (230 or 460 volts) specified by the procuring activity (see 6.2).

3.4.13.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. All motors (except axis drive motors), shall conform to the requirements of NEMA MG-1. Unless otherwise specified (see 6.2), motors shall be alternating-current (AC), 60 Hz, or direct current (DC) industrial-type motors with dripproof enclosures. Each motor control shall conform to the requirements of NEMA ICS-1 and NEMA ICS-3.

3.4.13.2 Worklight When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.13.3 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.



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3.5.1 Physical size limitations. Unless otherwise specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each grinding machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The grinding machine shall be capable of grinding work-workpieces in accordance with the test requirements in 4.8.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished, including the following:

1. One (1) set of wrenches normally furnished by the manufacturer.
2. Three (3) external grinding wheels complete with wheel mounts and balancing weights. Grading shall be for general purpose grinding.
3. Three (3) internal grinding wheels of the sizes necessary to grind the inside diameter capacity of the machine. The wheels shall be complete with the necessary quills, spindles, etc. to mount to the internal grinding attachment.
4. One (1) wheel balancing arbor and stand.
5. One (1) extra set of headstock and grinding wheel drive belts.
6. One (1) T-slot faceplate.
7. One (1) independent 4-jaw chuck.
8. One (1) set of drive dogs.
9. One (1) set of table splash guards.
10. Two (2) work centers, carbide tipped (one for the headstock and one for the rootstock).
11. Steady rest.
12. One (1) center rest.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).



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3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE number or Military or Federal specification number)
Type of filter

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, and machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapprovedby the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.



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4.6 Acceptance test. Acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The machine shall be operated at no load for not less than 2 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.4 through 4.8.4.1. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. The test part length shall be not less than one half of the stated capacity for the machine being tested. Workpiece diameter and material shall be at the contractor's option. The contractor shall demonstrate the machine's capability to step grind (at least 2 diameters), traverse grind and face grind a shoulder. All diameters shall be round within 0.0001 inch of a desired dimension with not less than 0.0001 inch taper per foot and a surface roughness of not more than 5 microinches. Axial runout of the ground shoulder shall not exceed 0.0001 inch total indicator reading (TIR) . All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Environmental Protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.6. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.8.6 Electromagmetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.



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4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains intended use, acquisition requirements, and information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grinding machines covered by this specification are intended for use in grinding environments where uniform tolerances for size, roundness and finish on a wide variety of materials is required. This type machine is versatile in that straight and tapered bores, external and internal diameters and rotary faces may be ground in one set-up. The designation of universal is derived from the machines ability to swivel the wheelhead and workhead.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound or metric) (see 3.2.1).

f. Digital readout system, it not required (see 3.2.1.2).

g. Special digital readout system features, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

j. Specify and fully describe additional environmental protection requirements, if required (see 3.2.6).



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k. Painting, if different (see 3.3.5).

1. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

m. Workhead center arrangement, if different (see 3.4.2).

n. Required internal taper of footstock, if different (see 3.4.3).

o. Power traverse of table, if different (see 3.4.4).

p. Swivel alignment device, if required (see 3.4.5.1).

q. Power rapid positioning, if required (see 3.4.6.3).

r. Automatic infeed cycle, if required (see 3.4.7).

s. Grinding wheels, if different (see 3.4.10).

t. Optional filtration system, if required (see 3.4.11).

u. Electrical requirements, if different (see 3.4.13).

v. Initial voltage, if different (see 3.4.13).

w. Motors, if different (see 3.4.13.1).

x. Worklight, if not required (see 3.4.13.2).

y. Hour meter, if required (see 3.4.13.3).

z. Size and capacity, if different (see 3.5).

aa. Specify physical size limitation, if required (see 3.5.1).

bb. Additional performance features, if required (see 3.7.1).

cc. Specify and fully describe optional equipment, if required (see 3.9).

dd. Specify applicable repair parts, if required (see 3.10).

ee. Lubrication plate, if required (see 3.11.1).

ff. Information to be provided on lubrication plate, if different (see 3.11.1).

gg. Nameplate, if different (see 3.11.2).

hh. Hydraulic plate, if required (see 3.11.3).



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ii. Hydraulic plate, if different (see 3.11.3).

jj. Inspection conditions, if different (see 4.3).

kk. First article inspection, if different (see 4.4).

11. Quality conformance inspection, if different (see 4.5).

mm. Acceptance test, if different (see 4.6).

nn. Additional performance tests, if required (see 4.8.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| N/A | DI-TMSS-80527 | Commercial Off-The-Shelf (COTS) Manuals | Require all data be in the English language |
| N/A | DI-TMSS-80528 | Supplemental Data For Commercial Off-The-Shelf (COTS) Manuals | Require all data be in the English language |

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products,



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who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Units (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

| MII-G-23823C | MIL-G-23823D | MIL-G-23823E |
|---|---|---|
| Size 1020 | Size 1 Class A | Size 1020 |
| Size 1024 | Size 1 Class B | Size 1024 |
| Size 1036 | Size 1 Class C | Size 1036 |
| Size 1236 | Size 2 Class A | Size 1236 |
| Size 1248 | Size 2 Class B | Size 1248 |
| Size 1272 | Size 2 Class C | Size 1272 |
| Size 1424 | Size 3 Class A | Size 1424 |
| Size 1436 | Size 3 Class B | Size 1436 |
| New size | Size 3 Class C | Size 1440 |
| Size 1448 | Size 3 Class D | Size 1448 |
| Size 1472 | Size 3 Class E | Size 1472 |
| Size 1496 | Size 3 Class F | Size 1496 |
| Size 1824 | Size 4 Class A | Size 1824 |
| Size 1836 | Size 4 Class B | Size 1836 |
| Size 1848 | Size 4 Class C | Size 1848 |
| Size 1872 | Size 4 Class D | Size 1872 |
| Size 1896 | Size 4 Class E | Size 1896 |
| New size | Size 5 Class A | Size 2424 |
| New size | Size 5 Class B | Size 2436 |
| New size | Size 5 Class C | Size 2448 |
| New size | Size 5 Class D | Size 2472 |
| New size | Size 5 Class E | Size 2496 |



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| | | |
|---|---|---|
| New size | Size 6 Class A | Size 3024 |
| New size | Size 6 Class B | Size 3036 |
| New size | Size 6 Class C | Size 3048 |
| New size | Size 6 Class D | Size 3072 |
| New size | Size 6 Class E | Size 3096 |

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

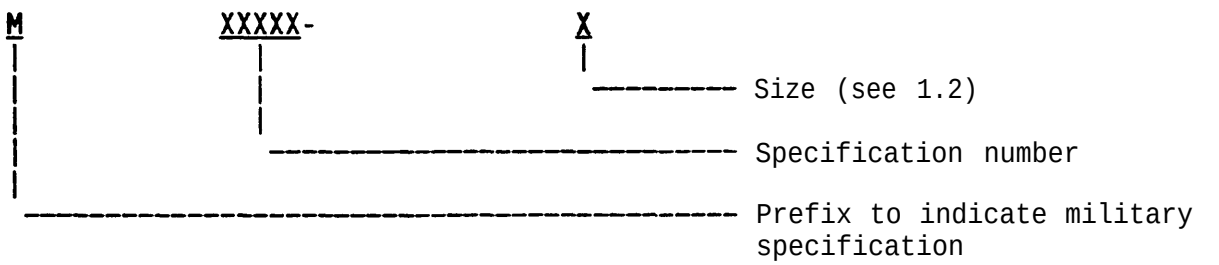
6.10 Hazardous materials minimization. The manufacturer should minimize the use, generation, or emission of materials known hazardous to the environment during the manufacturing process employed at his facilities. During the production of the end item, the manufacturer should comply with the applicable Federal, state, and local statutes in effect at the point of production which regulate the use of hazardous materials. Where practical, alternative materials or processes should be employed in lieu of known hazardous materials or processes to minimize the threat to the environment.

6.11 Training. Training required by the procuring activity should be provided as specified in the contract.

6.12 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.13 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.14 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:






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6.15 Subject term (keyword) listing.

Abrasive wheel
Automatic cycle
Footstock
Gage
Industrial equipment
Manufacturing
Precision machining
Production environment
Wheelhead



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TABLE I sizes and capacities.

| CHARACTERISTIC | REQUIREMENTS | | | | | |
|---|---|---|---|---|---|---|
| SIZE | 1020<br>1024<br>1036 | 1236<br>1248<br>1272 | 1424<br>1436<br>1440<br>1448<br>1472<br>1496 | 1824<br>1836<br>1848<br>1872<br>1896 | 2424<br>2436<br>2448<br>2496 | 3024<br>3036<br>3048<br>3072<br>3096 |
| Rated swing diameter, inches | 10 | 12 | 14 | 18 | 24 | 30 |
| Distance between centers, inches | 10 24 36 | 36 48 72 | 24 36 40<br>48 72 96 | 24 36 48<br>72 96 | 24 36 48<br>96 | 24 36 48<br>72 96 |
| Wheelhead: | | | | | | |
| Wheelhead swivel, degrees -clockwise | <-----45-----> | | | | | |
| counter-clockwise | <-----30-----> | | | | | |
| Automatic wheelhead infeed at table reversal, diameter reduction range, inches | <----- 0.0005 to 0.003 -----> | | | | | |
| Wheelhead movement per revolution of handwheel, inches, not more than | <----- 0.050 -----> | | | | | |
| Grinding wheel diameter, inches (standard) | 10 | 12 | 14 | 14 | 14 | 14 |
| Workhead: | | | | | | |
| Workhead swivel, degrees, counter-clockwise | <----- 90 -----> | | | | | |
| Workhead spindle speeds, number | <----- infinite -----> | | | | | |
| Workhead spindle speed range, RPM | 60 to 600 | 50 to 400 | 50 to 400 | 25 to 200 | 12 to 360 | 12 to 360 |
| Traversing table: | | | | | | |
| Table traverse speeds, number | <-----infinite-----> | | | | | |
| Table traverse speed range, inches per minute | 2 to 144 | 3 to 144 | 3 to 144 | 3 to 144 | 2 to 240 | 2 to 240 |
| Table traverse rates, per revolution of handwheel, range, inches | 0.200-0.920 | 0.200-0.920 | 0.200-0.920 | 0.200-0.730 | 0.100-0.500 | 0.100-0.500 |
| Motor horsepower: | | | | | | |
| Grinding wheelhead | 2 | 3 | 3 | 5 | 5 | 5 |
| Headstock | 1/4 | 1/4 | 1/4 | 1 | 2 | 2 |
| Internal grinding attachment | 1 | 1 | 1 | 2 | 2 | 2 |
| Coolant | 1/10 | 1/4 | 1/4 | 1/4 | 1/3 | 1/3 |
| Hydraulics | 3/4 | 3/4 | 3/4 | 3/4 | 2 | 2 |





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TABLE II. Alignment Tolerances

| Tests to be applied | Maximum Allowable deviation |
|---|---|
| **Bed and table:** | |
| Bed straight in longitudinal direction | 0.00025" per ft. |
| Bed flat or level in transverse direction, no twist permitted | 0.00025" per ft. |
| Guideways of headstock and footstock parallel with table movement | 0.00012" per ft. |
| **Headstock and spindle:** | |
| Axis of headstock parallel with table movement in vertical plane (rising toward free end of the mandrel) | 0 to 0.0005" per ft. |
| Axis of headstock parallel with table movement in horizontal plane (free end of mandrel inclined toward grinding wheel) | 0 to 0.0005" per ft. |
| *Center point for true running | 0.0002" TIR |
| *Taper of work spindle runs true: | |
| (1) Nearest spindle nose | 0.0002" TIR |
| (2) At a distance of 12 inches | 0.0008" TIR |
| **Footstock:** | |
| Sleeve parallel with table movement in vertical plane (front end rising only) | 0 to 0.002" per ft. |
| Sleeve parallel with table movement in horizontal plane (free end of mandrel inclined toward the grinding wheel) | 0 to 0.001" per ft. |
| Taper in sleeve parallel with table movement in vertical plane (free end of mandrel rising) | 0 to 0.0005" per ft. |
| Taper in sleeve parallel with table movement in horizontal plane (free end of mandrel inclined toward grinding wheel) | 0 to 0.0005 per ft. |
| Axis of centers (mandrel between centers) parallel with table movement in vertical plane (rising toward footstock end) | 0 to 0.0005" per ft. |
| **Grinding wheel spindle:** | |
| Taper of spindle running true | 0.0005" TIR |
| Spindle for axial slip measured at two points, displaced by 180 degrees | 0.001" TIR |
| Axis of wheel spindle parallel with table movement in vertical plane (rising toward free end of mandrel) | 0 to 0.0012" per ft. |
| Axis of wheel spindle parallel with table movement in horizontal plane (free end of mandrel inclined toward table) | 0 to 0.0012" per ft. |
| Axis of wheel spindle and headstock spindle at same height with respect to swivel plate | 0.004" |
| Rise and fall of wheel spindle in its infeed movement | 0.002" in total length of infeed motion |
| Infeed motion of wheelslide square with bed ways (upper table set for cylindrical grinding by means of a dial gage) | 0.001" on total length of infeed motion |

*Applicable to live spindle headstock only.



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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - AS
DLA - GS
Air Force - 84

User activities:
Army - AR
Navy - MC

Preparing activity:
DLA - IP

Project (3415-0190)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-G-23823E | 91 04 16 |

3. DOCUMENT TITLE
GRINDING MACHINES, CYLINDRICAL, EXTERNAL, CENTER TYPE UNIVERSAL, TRAVELING TABLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| Defense Industrial Plant Equipment Ctr<br>ATTN: DIPEC-SSM | (1) Commercial<br>901-775-4797 | (2) AUTOVON<br>683-4797 |
| c. ADDRESS (Include Zip Code)<br>2163 Airways Blvd.<br>Memphis, TN 38111-5051 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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